Manvansh Singh

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LinkedIn | GitHub | Portfolio

Education

Chandigarh Engineering College, CGC Landran — Landran, Punjab
B.Tech in Robotics & Artificial Intelligence — 07/2024 – Present

Technical Skills

Languages: Python, C/C++, JavaScript
Machine Learning: Regression, Classification (Random Forest, XGBoost), Anomaly Detection (Isolation Forest), Feature Engineering, Hyperparameter Tuning (GridSearchCV)
Deep Learning: PyTorch, CNNs, U-Net, Computer Vision
Frontend: HTML, CSS, React.js, Tailwind CSS
Backend & Database: Node.js, Express.js, FastAPI, MongoDB, MySQL, PostgreSQL
Python Libraries: NumPy, Pandas, Matplotlib, Seaborn, PyTorch, Scikit-learn, NiBabel, Plotly
Developer Tools: Git, GitHub, Postman, VS Code, Jupyter Lab, Vercel, Render

Experience

Devantra Innovations Pvt. Ltd. — Mohali

Co-Lead, Technical Department — Sep 2026 – Present

- Lead the technical department, coordinating tasks across the team, supporting teammates with problem-solving, and driving delivery.
- Led a client-facing web project end to end, owning technical execution, team coordination, and delivery; the project is live.

Software Developer — Jul 2026 – Sep 2026

- Led a team of 3 developers to design and deliver a full-stack Recruitment Drive Portal with dedicated Admin and Candidate portals covering the full recruitment lifecycle.
- Engineered 26 REST API endpoints using Node.js, Express.js, and PostgreSQL for authentication, recruitment drives, assessments, and result management; built 8+ pages and 7 reusable React components with role-based workflows.

Software Developer Intern — Jan 2026 – Jul 2026

- Architected and delivered 5 ERP modules by engineering 18 REST API endpoints with Node.js and Express.js, supporting daily operations across the platform.
- Implemented role-based authorization for Admin, Manager, and Member roles and built responsive interfaces across 10+ ERP pages, integrated with the backend.

Projects

Turbofan Predictive Maintenance System | *Python, FastAPI, XGBoost, Scikit-learn, React* — GitHub | Demo

- Engineered 46 time-series features (rolling stats, RUL labels) from NASA's C-MAPSS dataset (100 engines, 20K+ cycles) with engine-level splitting to prevent temporal data leakage
- Trained RF/XGBoost models (GridSearchCV) achieving R^2=0.79 (RUL), F1=0.87, Recall=0.93; Isolation Forest flagged failures 12 cycles out vs. 113 for normal readings

Dental CBCT Tooth Segmentation Pipeline | *PyTorch, U-Net, Medical Imaging* — GitHub | Demo

- Built a full preprocessing-to-postprocessing 2D U-Net pipeline (7.7M params) for 3D CBCT tooth segmentation, trained on 14,790 slices from 50 volumes on RTX 4050 (CUDA)
- Achieved 0.7479 Dice / 0.6948 IoU on held-out test set with custom PyTorch DataLoader and interactive 3D Plotly visualization of predictions

OceanEmbed | *PyTorch, CNN, FastAPI, React, Vite* — GitHub | Demo

- Built a CNN encoder–decoder in PyTorch reconstructing subsurface Bay of Bengal temperatures across 14 depth levels (0.5–700 m) on a 76×76 grid from SST, SSH, and SSS inputs
- Achieved 1.42°C RMSE on held-out GLORYS data, a 12.3% improvement over the 1.619°C climatology baseline; validated against CORA gridded objective analysis
- Deployed a full-stack FastAPI and React/Vite platform with interactive temperature heatmaps, depth-wise exploration, ocean-mask-aware statistics, and date-based reconstruction

Achievements & Certifications

200+ LeetCode Problems Solved – consistent DSA practice across various problems
Finalist – Smart India Hackathon (SIH), Internal Round, 2025
1st Prize – CodeQuest (College-level Coding Competition)
Industrial Training in AI/ML Using Python – 4-week program
Career Essentials in Generative AI – Microsoft and LinkedIn